

CMYK Color Blending with Prusa XL and More?

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Open-source code, all the photos, and some 3D scans will be available on [GitHub](#).

Abstract - The potential for multi-tool head 3D printers, such as the Prusa XL to achieve a broad color palette through primary color blending. By utilizing multi-tool heads that switch back and forth for a set number of layers, an optical illusion of color mixing is created. This research investigates the impact of various print parameters, including layer heights (with a focus on those under 0.10 mm), print speeds, and the number of tool heads, on print quality and color blending. Both Prusa Slicer software configurations and custom Python scripts were employed to manage the extrusion sequences.

Index Terms - CMYK, color bias, 3D printer, Python, G-Code, PLA

I. Introduction

3D printing is a fast-growing industry; it is a constantly changing field with numerous types of printers that have never been seen before. Originally viewed as a hobby, it is now being used in industrial rapid prototyping and products. From dental to military grade equipment, to now a household item, everyone uses a 3D printer. This paper uses a special multi tool-head printer from Prusa called Prusa XL to do color blending. The color blending with XL print at a very small layer height causes an illusion of color mixing. Will this be the new printing method standard for achieving controlled color printing as printers keep advancing with hardware and software? Current common printers have been designed to print filaments independently, so if your printer has 2 color filaments, it can only print those 2 colors. This blending method allows you to print those colors and the colors between them. This means consumers will no longer need to buy filament like light blue when it can be made with just white and blue; this saves consumers money and allows for customizable colors. There are already printers that allow you to create any color with ink coloring after the print, but these printers cost a lot more than the average printer.

II. Methodology

The testing of the prints was done with open lab equipment where students use the machines during the week. The main method of printing was using the Prusa Slicer, using PLA, and Prusa XL printer from 2T – 5T (tool-head). Speed preset in Prusa slicer was used which

include “0.20mm SPEED” and “0.20mm QUALITY”. The wipe tower at 70mm and brim size to 6mm were increased from default settings. The combined print layer height of each layered color that is mixed must be no bigger than 3mm; any bigger makes the color look more separated than blended. The tool-head switching in Prusa slicer is called “set extruder sequence” and the other method would be the custom Python script. We will talk about 2-5 color blending settings, and common errors within the software. The Python script includes an easier UI interface for users, hex-code color mixing preview to get an estimate of mixing result. The GitHub code includes a non-working color change over time that can be viewed in G-Code viewer but doesn’t work due to red screen from XL.

A. Common Errors

Always reslice or reload the object to update the layer sequence properly when adding a new object. Otherwise, the result may be incorrect, as shown in sample 5. Small bumps appear in the print when the calibration on the tool heads has not been done or needs to be done. Larger or complex geometrical shapes tend to fail or have greater color bias, like the lung sample.

B. Color Blending Prusa Slicer

To set the extruder sequence, it's located on the right corner of the print viewer with a gear symbol. This opens a window where tool-heads can be added or removed in a sequence. Let's say to make different types of light blue, use T1 (blue) and T2 (white). In the extruder sequence, do one of each layer. Make sure the layer change is at 1-layer for this blending to work, printing one layer of each color. Add one of each; this means it will print T1 first, then T2, which keeps repeating until the print is done. This only gets us one type of blue. Keep adding T1 or T2 to make it bluer or whiter. The sequence in Prusa Slicer would be T1, T1, T2.

1) *2+ Color Blending Prusa Slicer*: Let's add green with T3 to the sequence: T1, T3, T2; this will give us a light teal color. Be careful not to go over the recommended limit of a combined max layer height of 3mm. The limit is all 5 tool-heads.

2) *Stripes/Random*: In the same menu, there's a check box for random and/or manually input a stripe. Stripes could easily be done with the layer changed to 3 or more. This gives us 3 layers before changing to a new color.

C. Theory

The smallest single color layer can be 1.5mm, and it's possibly smaller, around 1.2mm. The combined layer height maximum is around 3mm but could be closer to 2mm. Here are all combinations with a 3mm combined layer height in mind.

1) *Two Tone*: The max number of combinations for two color mixing is also 13 at 0.05 mm layer height.

(T1, T2)

1, 1	1, 2	1, 3	1, 4	1, 5
2, 1	2, 3	2, 4	3, 1	3, 2
4, 1	4, 2	5, 1		

2) *Three Tone*: The RGB method has 19 new colors that can be mixed at 0.05mm with 3 tool-heads.

(R, G, B)

1, 1, 1	1, 1, 2	1, 2, 1	2, 1, 1	1, 1, 3
1, 2, 2	1, 3, 1	2, 1, 2	2, 2, 1	3, 1, 1
1, 1, 4	1, 2, 3	1, 3, 2	1, 4, 1	2, 1, 3
2, 3, 1	3, 1, 2	3, 2, 1	4, 1, 1	

3) *Four-Five Tone*: This requires 4/5 tool-heads; the 4th and 5th tools can be black or white, or any other color. The combination is just one variable (1 color) times two for black and white. The number combination is 15, but the total number of combinations is (15x2) 30. The total number of colors is 54, so with 5 filaments.

(C, M, Y, K)

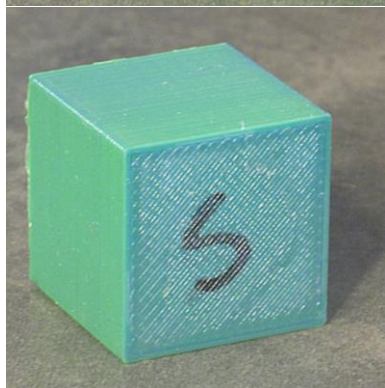
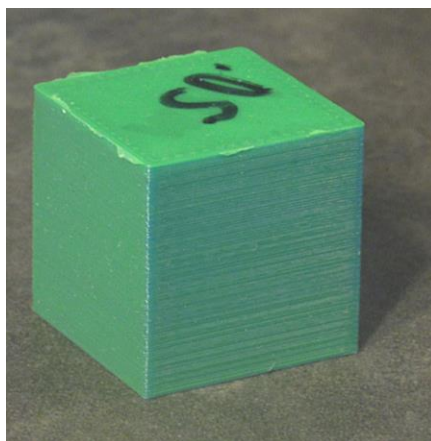
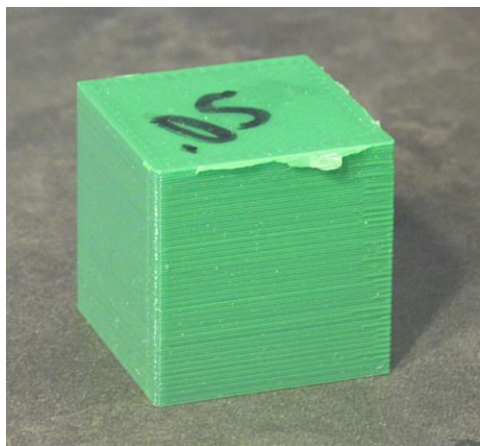
1, 1, 1, 1	2, 1, 1, 1	1, 2, 1, 1	1, 1, 2, 1	1, 1, 1, 2
3, 1, 1, 1	1, 3, 1, 1	1, 1, 3, 1	1, 1, 1, 3	2, 2, 1, 1
2, 1, 2, 1	2, 1, 1, 2	1, 2, 2, 1	1, 2, 1, 2	1, 1, 2, 2

III. Result

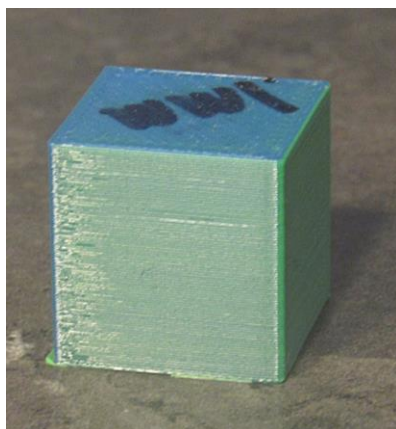
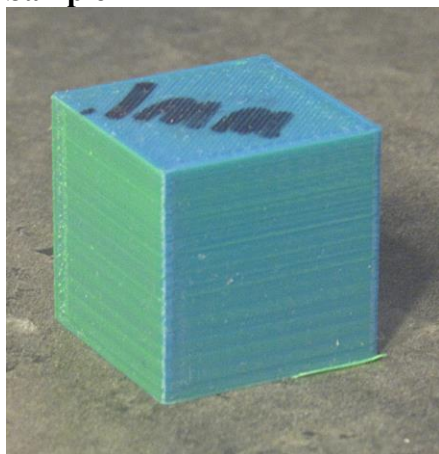
A. Photos

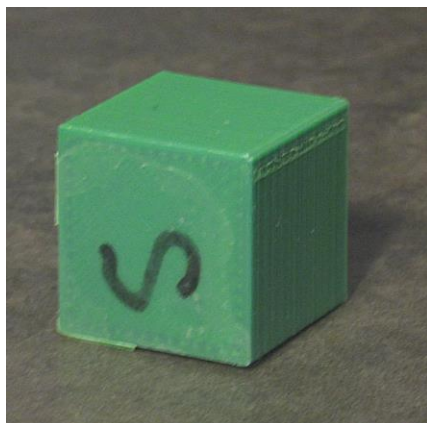
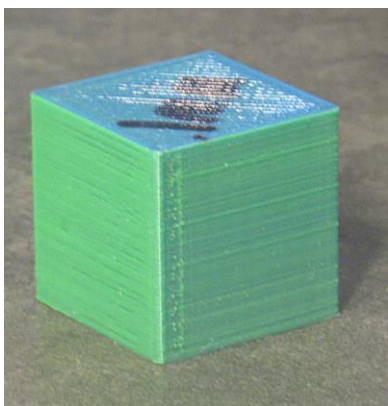
All the photos can be found on GitHub.

Sample 1

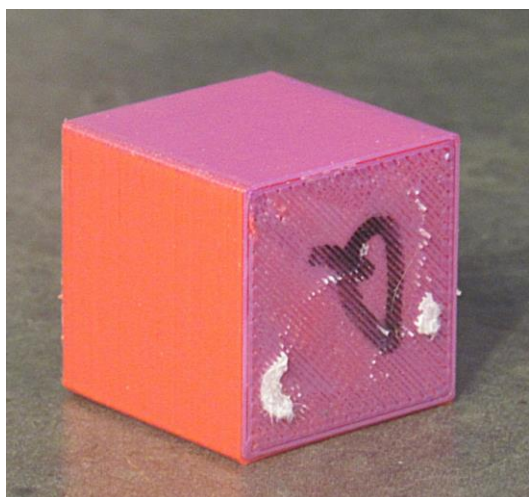
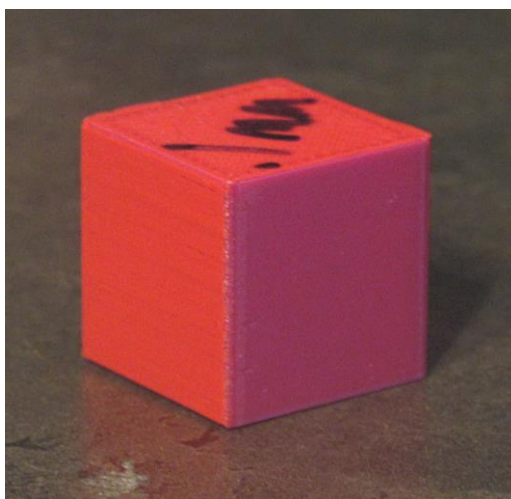
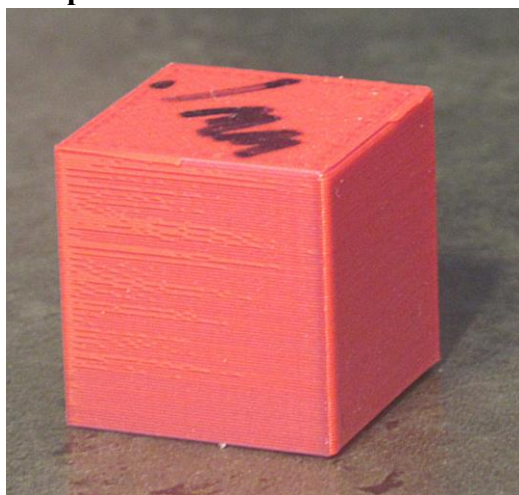


Sample 2

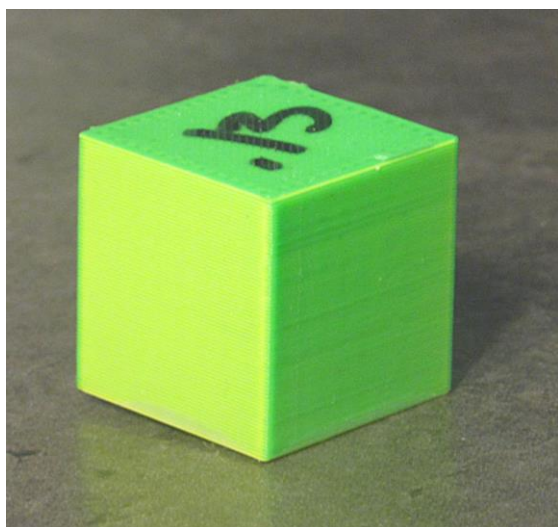
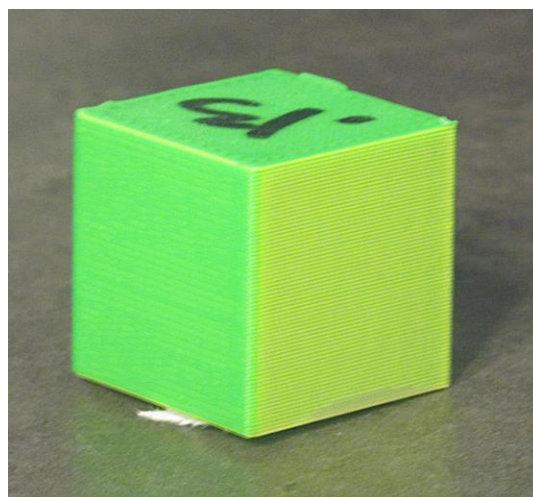
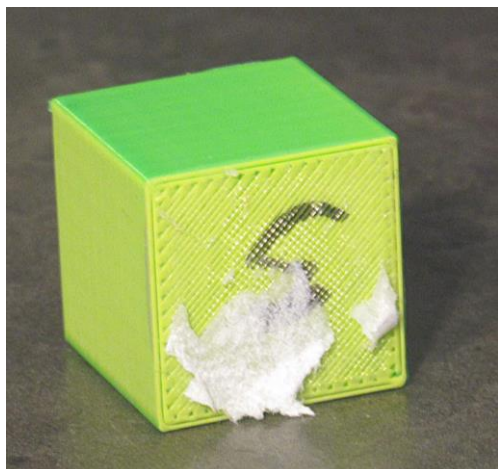




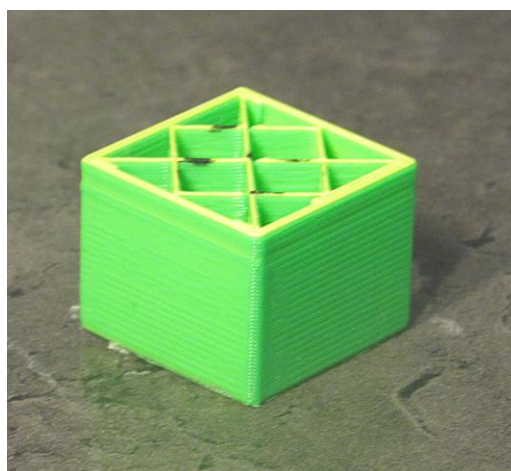
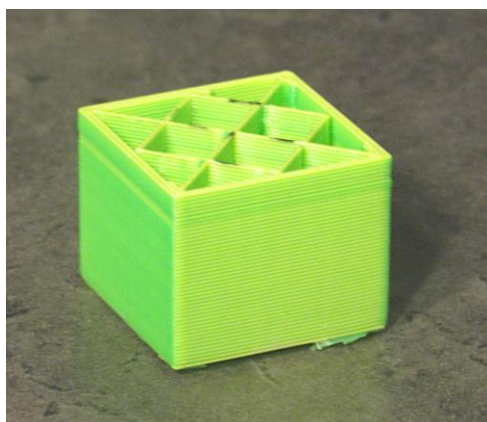
Sample 3



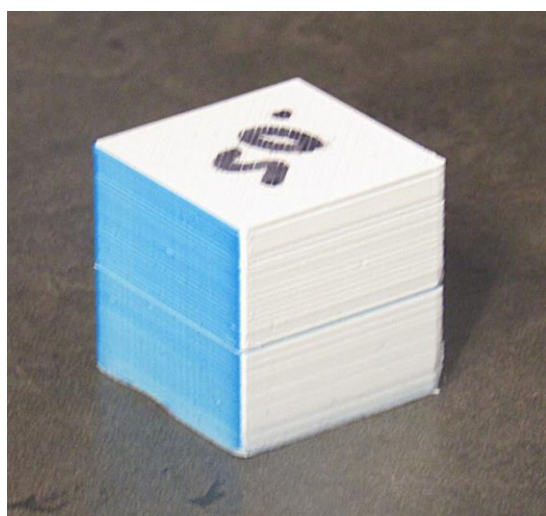
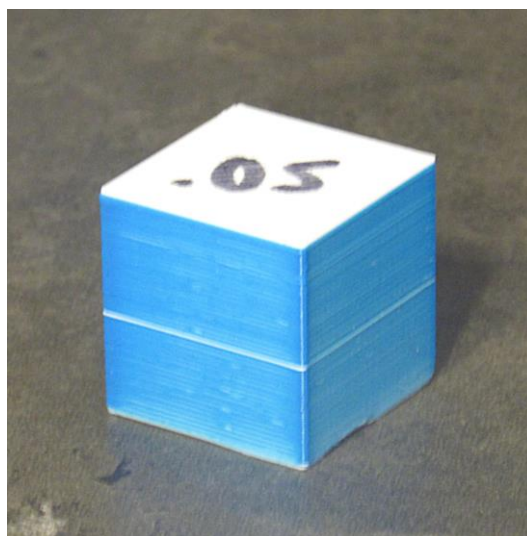
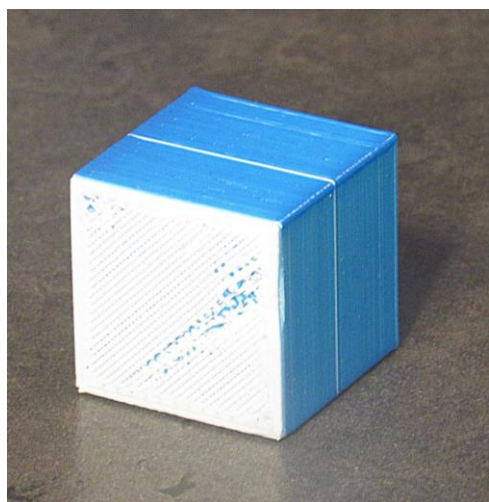
Sample 4



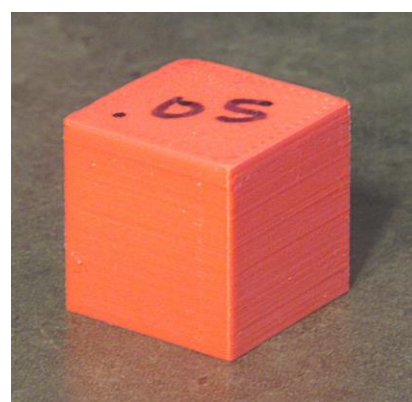
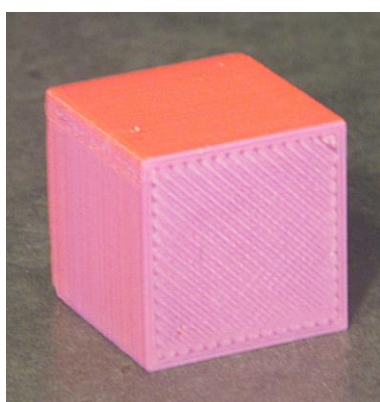
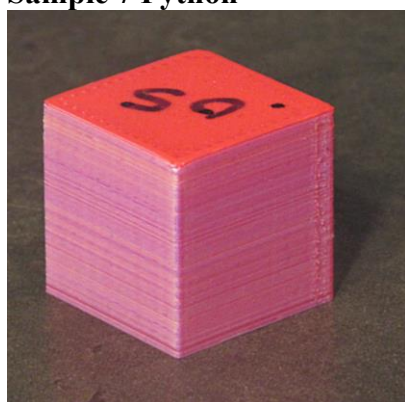
Sample 5



Sample 6



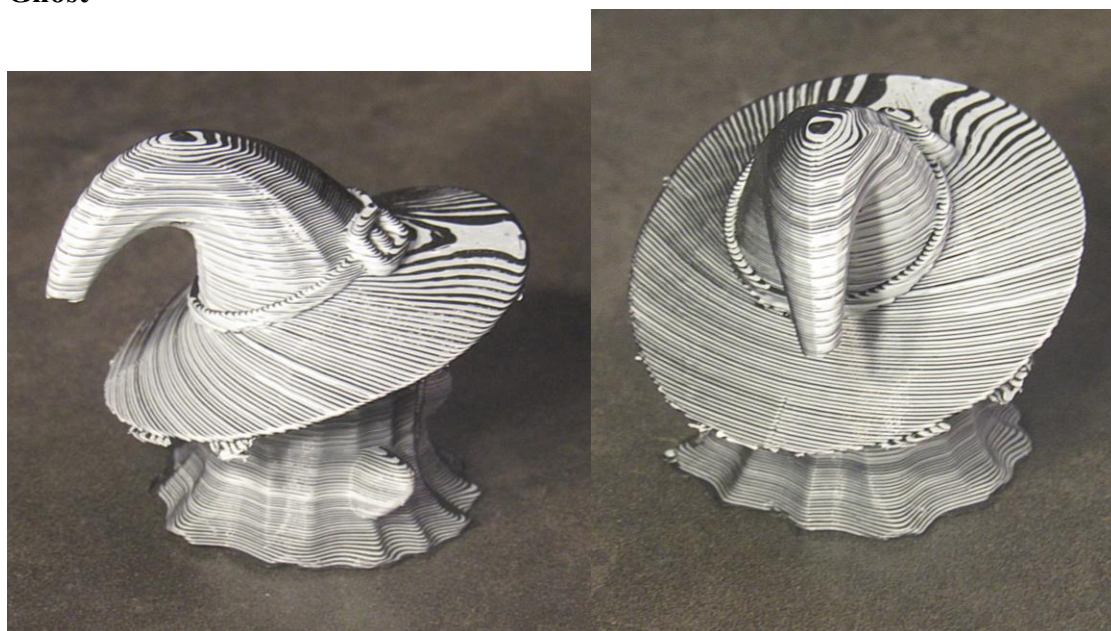
Sample 7 Python



Sample Heart



Ghost





Lung





B. Sample

Keeping the default settings (60mm for the tower and 3mm for the brim), the tower and the print often get knocked over, caused by the constant tool-head switching. Color bias occurs when one side is more color dominant, and the other side shows the other color. The print time increases from T1 to T2, but afterward, the time remains the same.

1) *Present Setting*: Before the actual documentation of this study, some prints weren't included. This explains why only one of the prints was QUALITY and the rest were SPEED. During this, QUALITY prints notably worse in color blending than in SPEED. The reason for this may be that SPEED doesn't allow the print to cool down fast enough, so when the next toolhead is ready, the printer is ready. The print is still warm and allows the colors to melt together. [1] This is why annealing could improve color blending by fusing the color print.

2) *Layer Height*: 0.05mm or smaller produces the best color quality print, because the eye has a harder time seeing the visible line difference in the print. Smaller layer height has a higher print success rate and fewer artifacts, which is why it is unknown (Sample 5).

3) *Size*: More complex and bigger prints tend to fail a lot more than smaller prints (Sample Lung).

4) *Color*: A print that uses white or black has a hard time retaining its secondary color. Increasing the secondary helps from black or white dominating the color.

Settings Key

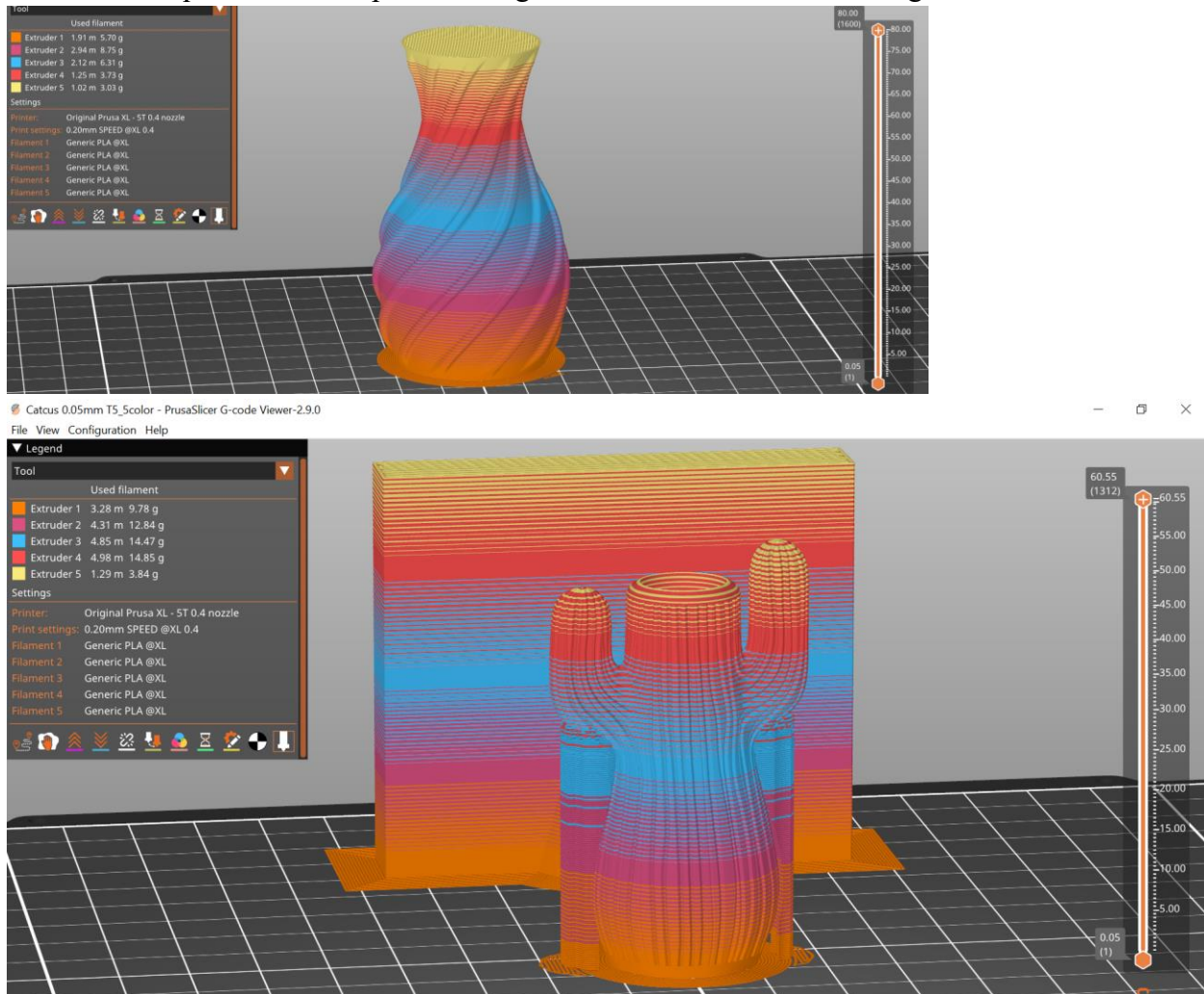
- **Sample**: The default cube model in Prusa Slicer.
- **Prusa Slicer Preset Setting**: S = Speed print, Q = Quality print
- **Colors**: G = Green, B = Blue, BL = Black, W = White, P = Purple, O = Orange, R = Red, Y = Yellow
- **Color Layer Number Example**: R:2, B:1 means 2 layers of red, then one layer of blue.
- **Toolhead Number Example**: T2 = 2 toolheads, T3 = 3 toolheads
- **Layer Height**: In mm
- **Python**: Denotes custom code; if not specified, it was made with Prusa Slicer.

Sample 1: S, T3, G:2, B:1, 0.05mm	Sample 2: S, T2, G:1, B:1, 0.10mm
Sample 3: Q, T2, P:1, R:1, 0.10mm	Sample 3: S, T3, Y:1, G:1, 0.13mm
Sample 5: S, T2, Y:1, G:1, 0.15mm FAIL	Sample 6: S, T2, B:1, W:1, 0.05mm
Sample 7: S, T3, R:2, P:1, W:1, 0.05mm Python	Lungs: S, T3, R:2, P:1, W:1, 0.05mm FAIL
Heart: S, T2, W:1, R:1, 0.05mm	Ghost: S, T2, B:1, W:1, 0.05mm
Dino: S, T2, O:1, R:1, 0.05mm	Rabbit: S, T2, Y:1, W:1, 0.10mm

III. Discussion

A. Changing Color Over Time

A Python script was written to change color over time, but this did not work during testing. The error was that it didn't pick up the tool head during print, although the G-Code viewer showed it changing correctly. The script was designed to print one whole color, then slowly add the next color layer by layer until that color was solid, then repeat the process with the next color. Allowing the creation of a custom ratio that changes over time. One of the G-Code made a quarter of the top with a single color, and then the rest changed.



B. Future

1) *Software and System:* Imagine a system where a 3D filament has a hex-color code on it, and the slicer can visually predict its color with a 3D model. A color tool could also allow users to choose a color and paint on each layer. The printer of the future will perform this task better than current models, with enhanced speed and accuracy in tool-head switching.

2) *Filament:* Mixing primary color filaments allows customers to save money by not having to buy every specific color and reduces material waste. PLA is recyclable, so the wipe tower and all extra materials can be recycled. This research has gotten feedback "Maybe you could even explore not only swapping colors but materials, to get combined mechanical properties.". This could result in users making custom materials or having different material

sections in their prints, like external walls being different from inner walls. Annealing [1] could be a post-processing technique that melts the print. This could result in less visible lines and a better color mix.

IV. Conclusion

The results indicate the potential for color blending. Regular recalibration of the machine, along with limiting its usage to specific tasks, is likely to yield more reliable results. Further research is essential to achieve improved color outcomes, and techniques such as annealing may offer beneficial solutions. Additionally, blending materials could enhance the customization of mechanical properties in printed objects. Overall, while some success has been documented, there is a clear need for continued exploration and refinement in these areas.

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Reference

- [1] Kočí, Jakub. “How to Improve Your 3D Prints with Annealing.” *Original Prusa 3D Printers*, 17 Dec. 2019, blog.prusa3d.com/how-to-improve-your-3d-prints-with-annealing_31088/.